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Lecture #142 Immunogenetics, Lymphocyte Activation, Immune Regulation and Tolerance Part 2

Bradley Hickey, MD, PhD, MPH, MBA, MA, FACOEM
Assistant Professor, Department of Clinical Medicine
NYIT COM at Arkansas State University
Email: bhicke03@nyit.edu

Lecture Orientation:
From Recognition to Response

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- Antigen recognition is only the beginning
- This lecture focuses on events *after* activation
- Expansion, shaping, restraint, and resolution determine outcomes
- Downstream control defines protection versus pathology

Learning Objectives and Conceptual Roadmap

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- Understand how activation becomes functional immunity
- Explain expansion, differentiation, memory, and regulation
- Emphasize *why* regulation is necessary
- Core theme: power requires restraint

Clonal Selection as a Foundational Principle

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- Only antigen-specific lymphocytes are selected
- Ensures specificity
- Creates a quantitative problem
- Antigen-specific cells are rare

Why Clonal Expansion Is Essential

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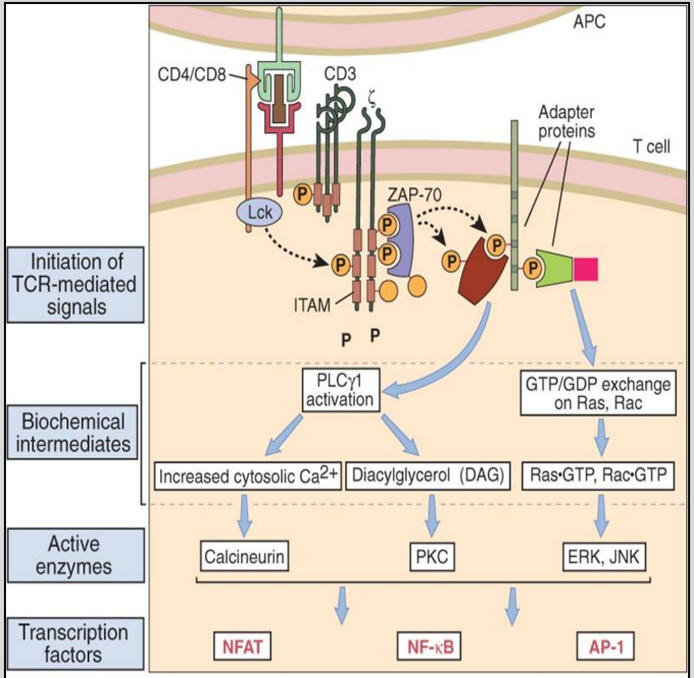
- Recognition alone cannot control disease
- Expansion generates sufficient effector numbers
- Amplifies rare specific cells
- Expansion is powerful but dangerous

Signals That Support Expansion

- Antigen recognition is insufficient
- Costimulation promotes survival and proliferation
- Cytokines reinforce expansion decisions
- Context determines whether expansion occurs



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Differentiation: Giving Cells Functional Identity

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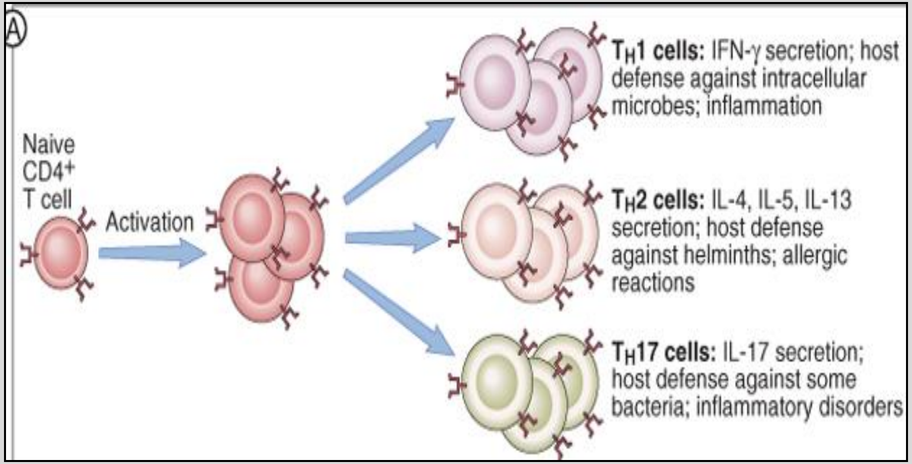
- Proliferating cells do not remain identical
- Differentiation assigns immune roles
- Cytotoxic
- Helper
- Coordinating functions



Abbas and Lichtman, Basic Immunology

Functional Diversity of Effector Cells

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- Effector cells perform immune work
- Different programs suit different threats
- Responses are tailored, not uniform
- Diversity improves effectiveness and safety

Memory as a Deliberate Immune Strategy

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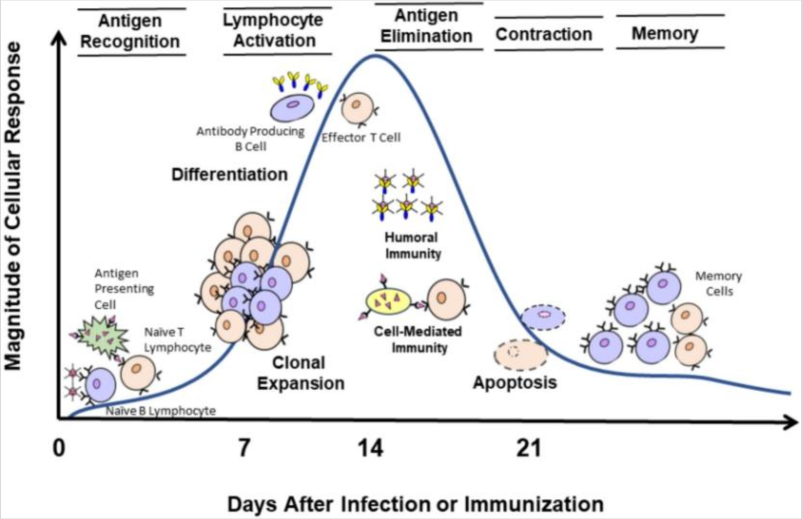
- Some activated cells become long-lived
- Memory enables faster secondary responses
- Memory formation is intentional
- Defining feature of adaptive immunity



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The Need for Controlled Contraction

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- Immune responses must end
- Persistent activation causes damage
- Contraction restores balance
- Failure leads to chronic inflammation

Regulation and Tolerance Mechanisms

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- Regulation limits immune damage
- Tolerance prevents responses to self
- Essential for immune safety
- Prepares for deeper discussion

Transition From Activation to Expansion

- Decision phase has ended
- Immune system commits resources
- Size and intensity must be controlled
- Errors cause weak or damaging responses

Clonal Expansion as a Quantitative Solution

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- Antigen-specific cells are initially rare
- Expansion creates large identical populations
- Converts recognition into clinical impact
- Precision without expansion is ineffective

Metabolic Demands of Rapid Proliferation

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- Resting lymphocytes are metabolically quiet
- Activation drives metabolic reprogramming
- Increased glucose and biosynthesis required
- Restricted to properly activated cells

Cytokines as Drivers of Expansion

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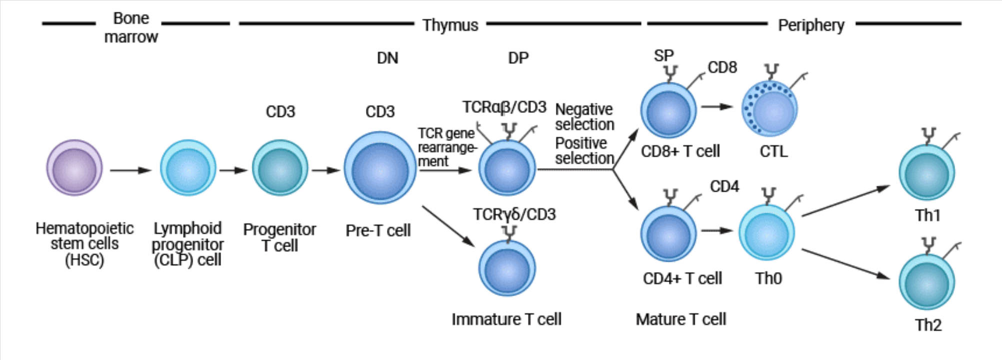
- Cytokines prevent apoptosis
- Promote cell cycle progression
- Scale expansion to threat severity
- Integrate environmental context

Effector Commitment During Expansion

- Expansion and differentiation occur together
- Early signals shape effector fate
- Function is determined early
- Context guides immune behavior



T-Cell Development. Reprinted from CUSABIO, https://www.cusabio.com/c-21194.html

CD8 T-Cell Effector Differentiation

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- CD8 cells become cytotoxic effectors
- Acquire killing machinery
- Differentiation is tightly regulated
- Prevents collateral tissue damage

CD4 T-Cell Differentiation Pathways

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- CD4 cells differentiate into helper subsets
- Cytokine signals guide fate
- Tailors immune responses
- Prevents oversimplification

Why Effector Differentiation Must Be Regulated

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- Effector programs affect host tissues
- Excess causes damage
- Insufficient responses fail
- Regulation balances effectiveness and safety

Linking Expansion to Immune Function

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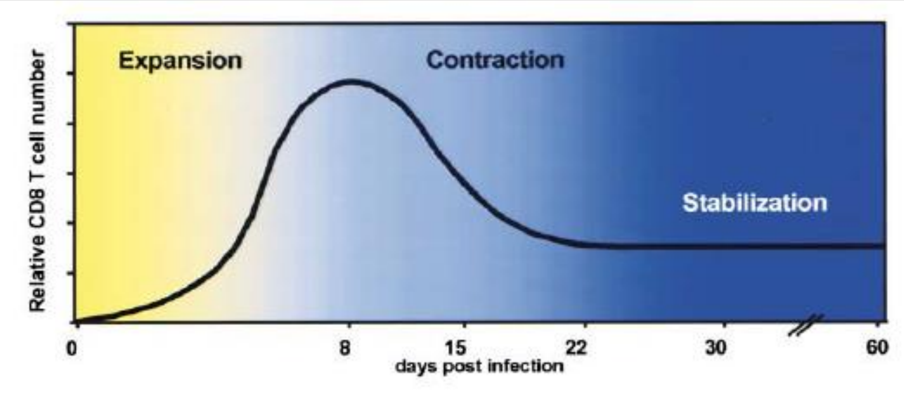
- Population size matters
- Population composition matters
- Expansion and function are inseparable
- Coordination determines outcome



Hemby, S., et al. Molecular and Functional Profiling of Memory CD8 T Cell Differentiation. Cell: 111(6):837-51.

Preparing for Memory and Contraction

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- Not all expanded cells persist
- Early signals influence fate
- Some become memory
- Others are eliminated

Expansion Is Followed by Contraction

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- Expansion cannot continue indefinitely
- Effector cells undergo apoptosis
- Restores immune homeostasis
- Active, regulated process

Why Contraction Is Not Immune Failure

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- Elimination signals success
- Prevents sustained tissue injury
- Responses must be self-limited
- Resolution is intentional

Memory Cell Survival During Contraction

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- Small subset survives contraction
- Memory cells are metabolically distinct
- Optimized for longevity
- Survival is regulated

Effector Versus Memory Cell Fates

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- Same precursors, different outcomes
- Effector: immediate function
- Memory: rapid recall and persistence
- Dual strategy ensures protection

Signals That Promote Memory Formation

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- Cytokine environment influences fate
- Antigen strength and duration matter
- Gene and metabolic programs shift
- Memory is not random

Functional Advantages of Memory Cells

- Faster activation
- Reduced costimulation requirement
- Stronger secondary responses
- Must still be regulated

Homeostatic Maintenance of Memory Pools

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- Memory persists without antigen
- Maintained by survival cytokines
- Avoids chronic inflammation
- Ongoing regulation in health

Immune Resolution as an Active Process

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- Resolution requires coordination
- Effector loss, memory preservation, repair
- Poor coordination causes disease
- Shutdown is complex

Clinical Implications of Failed Contraction

- Persistent effectors drive pathology
- Seen in chronic inflammatory disease
- Therapies aim to restore regulation
- Control loops matter clinically

Transition to Tolerance and Regulation

- Contraction leads into tolerance
- Prevents attack on self
- Failures cause autoimmunity
- Shift to regulation focus

Why Tolerance Is Required

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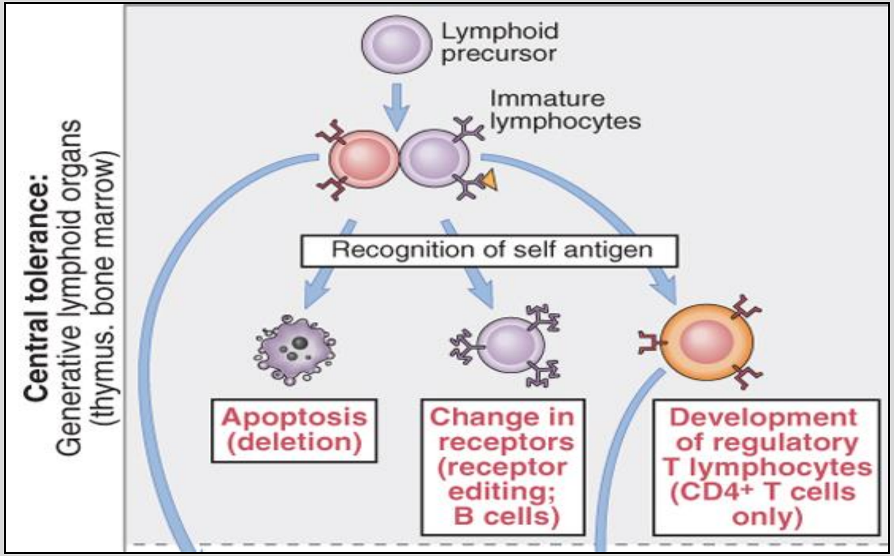
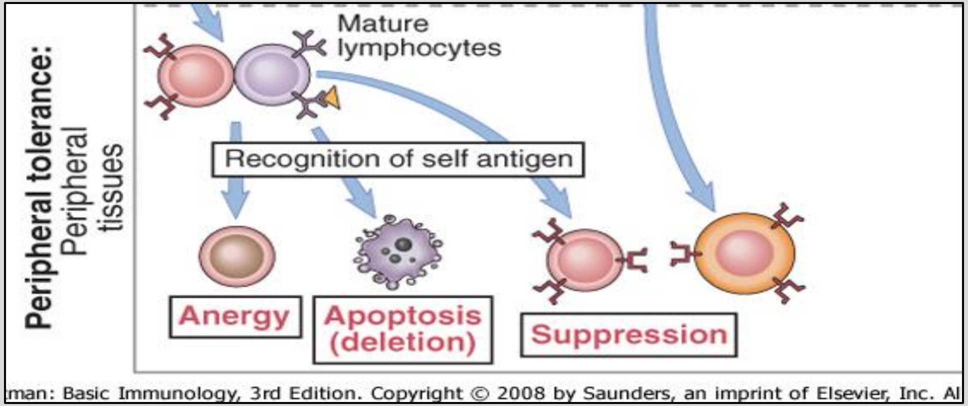
- Receptor specificity risks self-reactivity
- Tolerance controls dangerous clones
- Essential for survival
- Failure underlies autoimmune disease



Abbas and Lichtman, Basic Immunology

Central Versus Peripheral Tolerance

- Multiple tolerance layers
- Central: development
- Peripheral: mature lymphocytes
- Redundancy provides safety



Abbas and Lichtman, Basic Immunology

Central Tolerance in T-Cell Development

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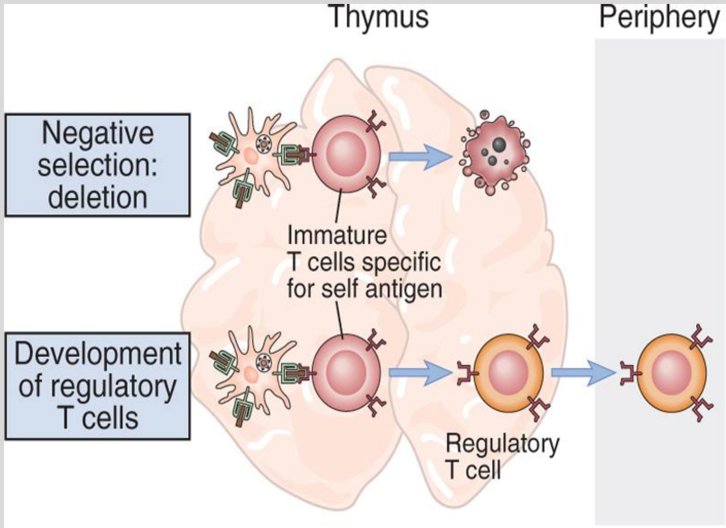
- Occurs in thymus
- Ensures self-MHC recognition
- Eliminates nonfunctional cells
- Balances recognition and risk



Negative Selection and Self-Reactivity

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- Strong self-binding eliminated
- Prevents high-risk clones
- Incomplete by necessity
- Peripheral tolerance required

Peripheral Tolerance: An Ongoing Requirement

- Controls mature lymphocytes
- Self antigens encountered lifelong
- Mechanisms include anergy, deletion, regulation
- Dynamic process

Anergy as a Tolerance Mechanism

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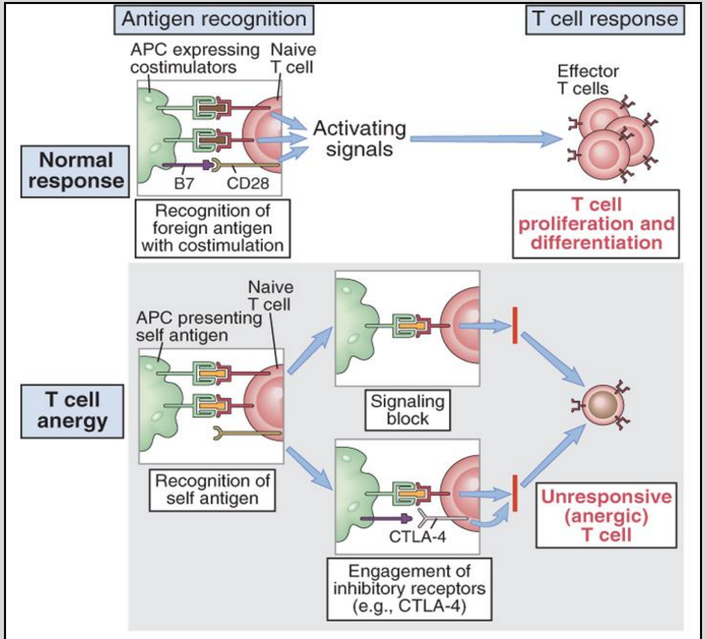
- Functional inactivation
- Occurs without costimulation
- Cells survive but do not respond
- Reinforces context dependence



Regulatory T Cells and Immune Control

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- Actively suppress responses
- Maintain immune homeostasis
- Control APCs and effectors
- Essential for stability

FOXP3 and Regulatory T-Cell Identity

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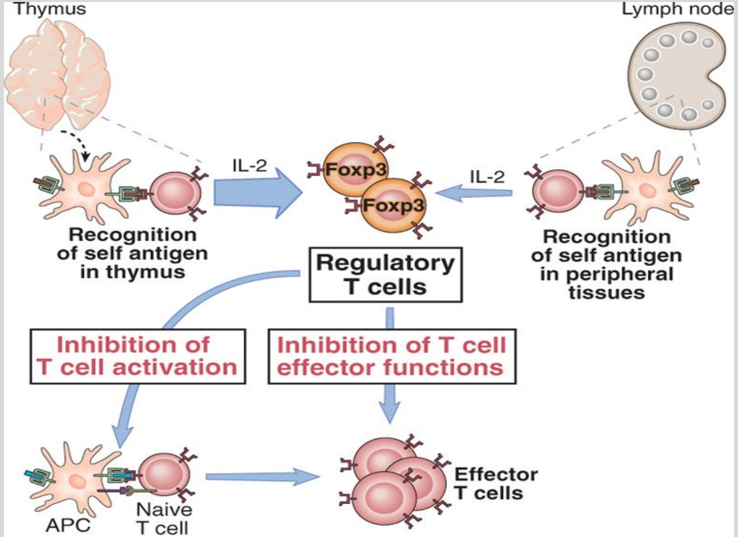
- FOXP3 defines regulatory program
- Required for suppressive function
- Loss causes immune dysregulation
- Regulation is active

The ***FOXP3*** gene provides instructions for producing the forkhead box P3 **(FOXP3)** protein. The FOXP3 protein attaches (binds) to specific regions of DNA & helps control the activity of genes that are involved in regulating the immune system.



Consequences of Failed Regulation

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- Autoimmunity and inflammation
- Breakdown of tolerance pathways
- Explains disease mechanisms
- Informs therapy

Integration of Tolerance With Immune Responses

- Regulation shapes, not silences immunity
- Contextual control
- Balance is central theme
- Sets stage for transplantation

B-Cell Tolerance: Parallels and Differences

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- B cells also risk self-reactivity
- Distinct tolerance mechanisms
- Complements T-cell tolerance
- Integrated protection

Central B-Cell Tolerance in the Bone Marrow

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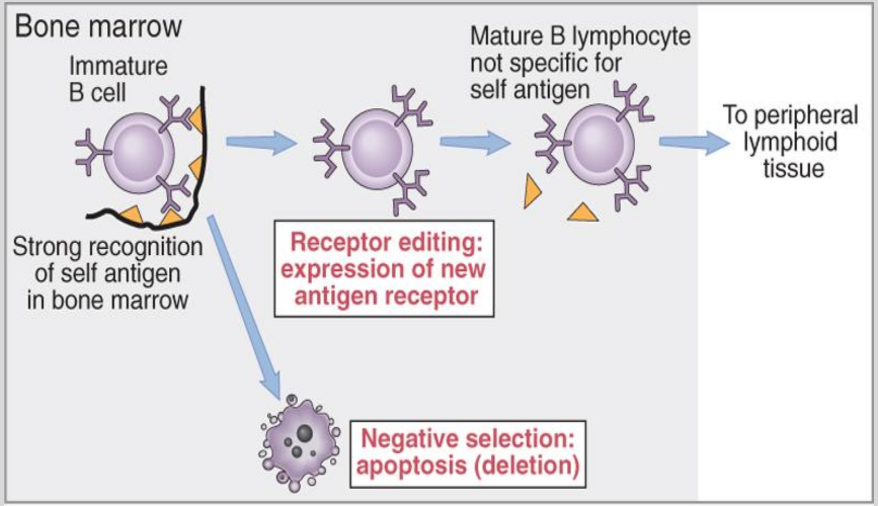
- Occurs during development
- Eliminates or alters self-reactive cells
- Additional flexibility vs T cells
- Layered safety



Receptor Editing as a Unique B-Cell Strategy

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- Alters antigen specificity
- Preserves useful cells
- Reduces self-reactivity
- Expands tolerance capacity

Peripheral B-Cell Tolerance Mechanisms

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- Controls escaped self-reactive cells
- Includes anergy and deletion
- Requires T-cell help
- Lifelong control

Why T-Cell Help Is Required for Antibody Responses

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- Acts as safety checkpoint
- Prevents autoantibody production
- Integrates tolerance layers
- Loss contributes to disease

Introduction to Transplantation Immunology

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- Foreign tissue introduced
- Persistent non-self
- Applies tolerance principles
- Clinically important model

Allorecognition: Direct and Indirect Pathways

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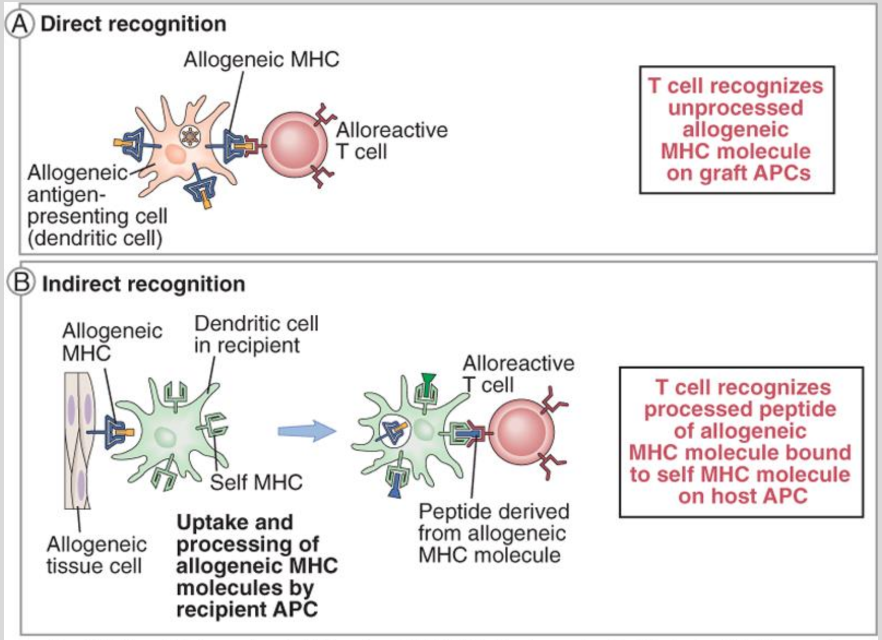
- Direct recognition of donor MHC
- Indirect recognition via processing
- Both activate recipient T cells
- Explain rejection mechanisms



Types of Transplant Rejection

- Hyperacute
- Acute
- Chronic
- Timing reflects mechanism

Immunosuppression and Its Tradeoffs

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- Prevents rejection
- Increases infection risk
- Balancing act
- Reflects immune principles

Integration of Tolerance and Clinical Practice

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- Central to autoimmunity and transplantation
- Mechanism-based reasoning
- Scales across specialties
- Immunology as framework

Key Concepts

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- Activation → expansion → differentiation → regulation
- Tolerance prevents damage
- Memory preserves protection
- Balance is essential

Closing and Forward Look

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- Completes adaptive response sequence
- Concepts recur in future lectures
- Immunology is cumulative

Summary Slide

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- **Summary Point 1: Adaptive Immunity Requires Controlled Amplification**

- Antigen recognition alone is insufficient to control infection
- Clonal expansion converts rare antigen-specific lymphocytes into effective populations
- Differentiation assigns specialized effector and helper functions
- Expansion without regulation leads to host tissue damage

- **Summary Point 2: Regulation and Tolerance Are Essential for Immune Safety**

- Contraction actively terminates immune responses after pathogen control
- Memory preserves protection while minimizing ongoing activation
- Central and peripheral tolerance prevent immune responses to self
- Regulatory mechanisms balance protection with long-term tissue integrity

- Core References:
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